Dictionaries and records

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Python readiness portfolio

THE CONNECTED WEEK

Teach field identity, value and validity as separate checks.

Supply complete named records, preserve wrong-key errors and clarify None versus absence before assessment.

Keep one habit: Ask whether the key exists, which value it holds and whether that value meets the task's schema.

MAIN LESSON · SECTION 1

Prepare the connected bridge lesson

> Check prior exposure and keep execution practice distinct from paper reasoning.

Goal and main plan

Primary target: Read and update named **dictionary** fields

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one **record**ed attempt; they are not new **evidence**.

Retrieval answer guidance

Retrieval and prerequisite guidance

Without opening the old lesson, explain one key idea from Return values and local state. Give an example and a mistake that the idea helps you avoid.

Look for: Use a returned value in a later calculation

Reference example, not the only acceptable wording: result is 9 and doubled is 18. Nothing is displayed by these assignments or the returning function. print(doubled) would display 18.

Lists and indexing

Without opening the old lesson, explain one key idea from Lists and indexing. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish a list index from the value stored at that position

Reference example, not the only acceptable wording: Length 3; indices 0, 1, 2; counts[1] is 7. The equal values occupy different positions, indices 0 and 2.

Boolean conditions

Without opening the old lesson, explain one key idea from Boolean conditions. Give an example and a mistake that the idea helps you avoid.

Look for: Translate stated requirements into comparisons

Reference example, not the only acceptable wording: (minutes <= 12) and quiet. At most includes equality, so <= is required; the nonnegative integer and Boolean **input** contract still applies.

Teach Week 10 as a two-session investigation using fictional data. A fictional cupboard **record** has a name, capacity and used count. A list asks us to remember positions. Named fields make the intended meaning visible.

Our schema requires name as text and capacity and used as nonnegative integers, with used no greater than capacity. Assess the three stated outcomes through the independent task, not by copying the worked result.

Materials and readiness

Python 3 editor with a fresh local practice file

Paper trace table and the student workbook

Fictional **input record**s from this lesson; no real student identifiers

5; the caller may store or print it.

End goal

Trace dictionaries and **record**s and explain the changed program before running it.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Return values and local state

Outcomes

Read and update named **dictionary** fields

Connect a **record** to a function call

Distinguish missing fields from zero values

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach named fields and unavailable values without conflating them

> Supply the Red record before asking for its update.

Read schema, Blue update and all record inputs

A fictional cupboard **record** has a name, capacity and used count. A list asks us to remember positions. Named fields make the intended meaning visible.

Our schema requires name as text and capacity and used as nonnegative integers, with used no greater than capacity.

Square brackets look up the value for a key. The quoted key "used" is text; its value is an integer. Assigning to an existing key replaces that field. Other fields remain the same.

Dictionary syntax does not enforce our schema: the programmer must check it.

Separate complete fictional record	Initial data and intended use
Blue · Main script	{"name":"Blue","capacity":12,"used":4}; update used to 6, remaining 8 then 6
Red · W1/W2 and error practice	{"name":"Red","capacity":9,"used":2}; W2 changes used to 9, remaining 0
Green · Earlier additional transfer	Its own task supplies capacity 15/used 6 then used 10; keep separate
Schema	name text, capacity/used nonnegative integers and used<=capacity. Example arithmetic does not enforce it.

Model complete Red practice

```
record = {"name": "Red", "capacity": 9, "used": 2}
def remaining(capacity, used):
    return capacity - used

record["used"] = 9
print(remaining(record["capacity"], record["used"]))
print(record["capacity"])
```

Read the shelf zero/missing example

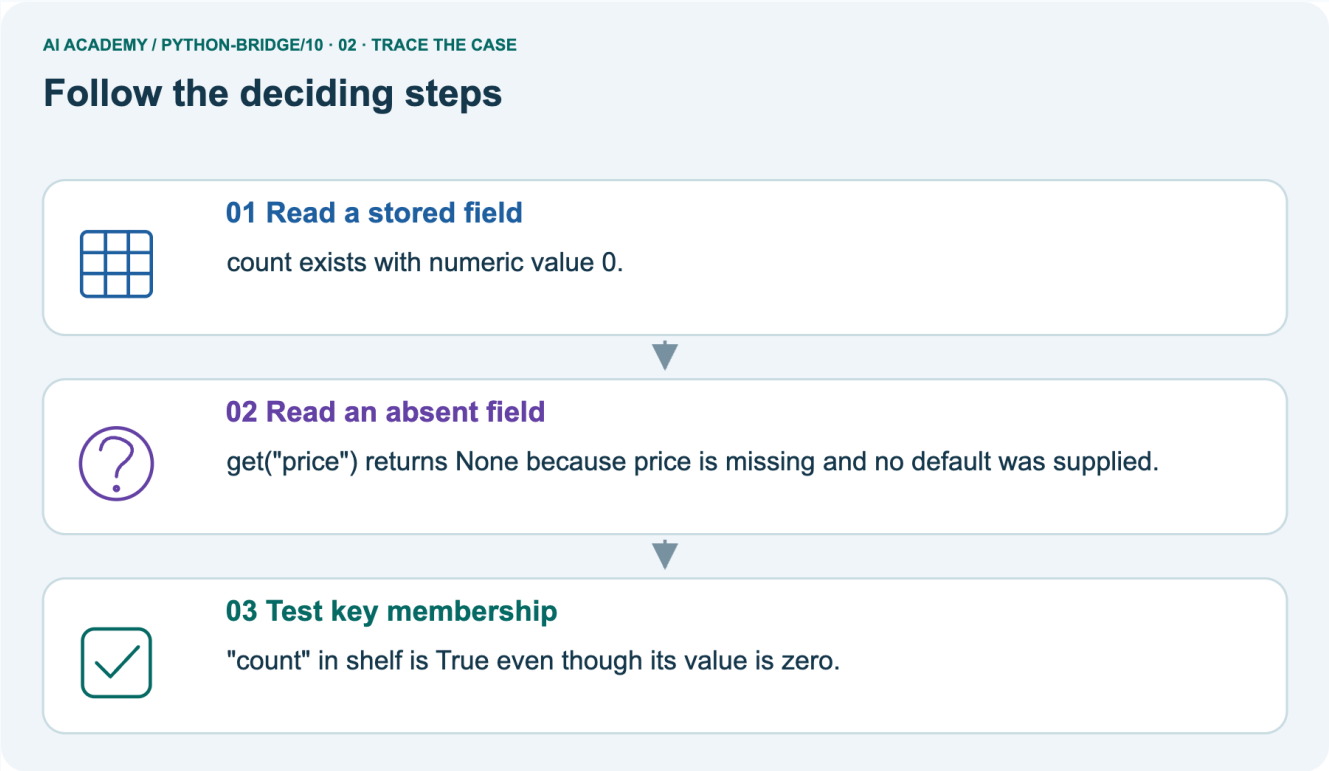
A zero field is still present. A **dictionary record** describes a shelf. The program distinguishes missing data from an actual zero.

Python 3: shelf={"count":0,"name":"Atlas"}; print(shelf["count"]); print(shelf.get("price")); print("count" in shelf).

For the pictured shelf, count is present with 0 and price is absent. get without a default returns None for absence, while membership checks whether the key exists. A present key can also explicitly hold None: get then gives None but membership is True. Do not call get alone a presence test.

The selected describe function uses its own narrow policy that absent/None count is unavailable and returns the label missing; supplied numeric 0 must remain a value. It does not validate arbitrary record types or all possible counts.

Trace lookup and membership



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Read a stored field	count exists with numeric value 0.
2. Read an absent field	get("price") returns None because price is missing and no default was supplied.
3. Test key membership	"count" in shelf is True even though its value is zero.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Dictionary keys name fields. Presence is different from truthiness of a value; a measured zero should not be discarded as if the field were absent.

```
shelf = {"count": 0, "name": "Atlas"}
print(shelf["count"])
print(shelf.get("price"))
print("count" in shelf)
```

Predict added zero price

Change: Add the field price with value 0.

Prediction: The price lookup now returns 0 instead of None.

Reason: A **record**ed zero is a present value, whereas the earlier lookup had no field.

State explicit None policy limits

None is also a possible explicitly stored value; real **record** contracts must define how missing and null values are represented.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

▼ Additional guided key

Spaces are 0. A present used field with value 0 is different from an absent field. Exit: No. The used count exceeds capacity. Successful arithmetic does not validate the **record**.

▼ Earlier wording, alternatives and picture transcripts

Term	Meaning
Dictionary	A mapping from unique keys to values
Key	A name used to look up a value
Record	Related fields describing one item
KeyError	The error from looking up an absent dictionary key

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
Read used	4	12 - 4 = 8
Update used	6	12 - 6 = 6

Read each row from left to right and connect the evidence to the stated rule.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

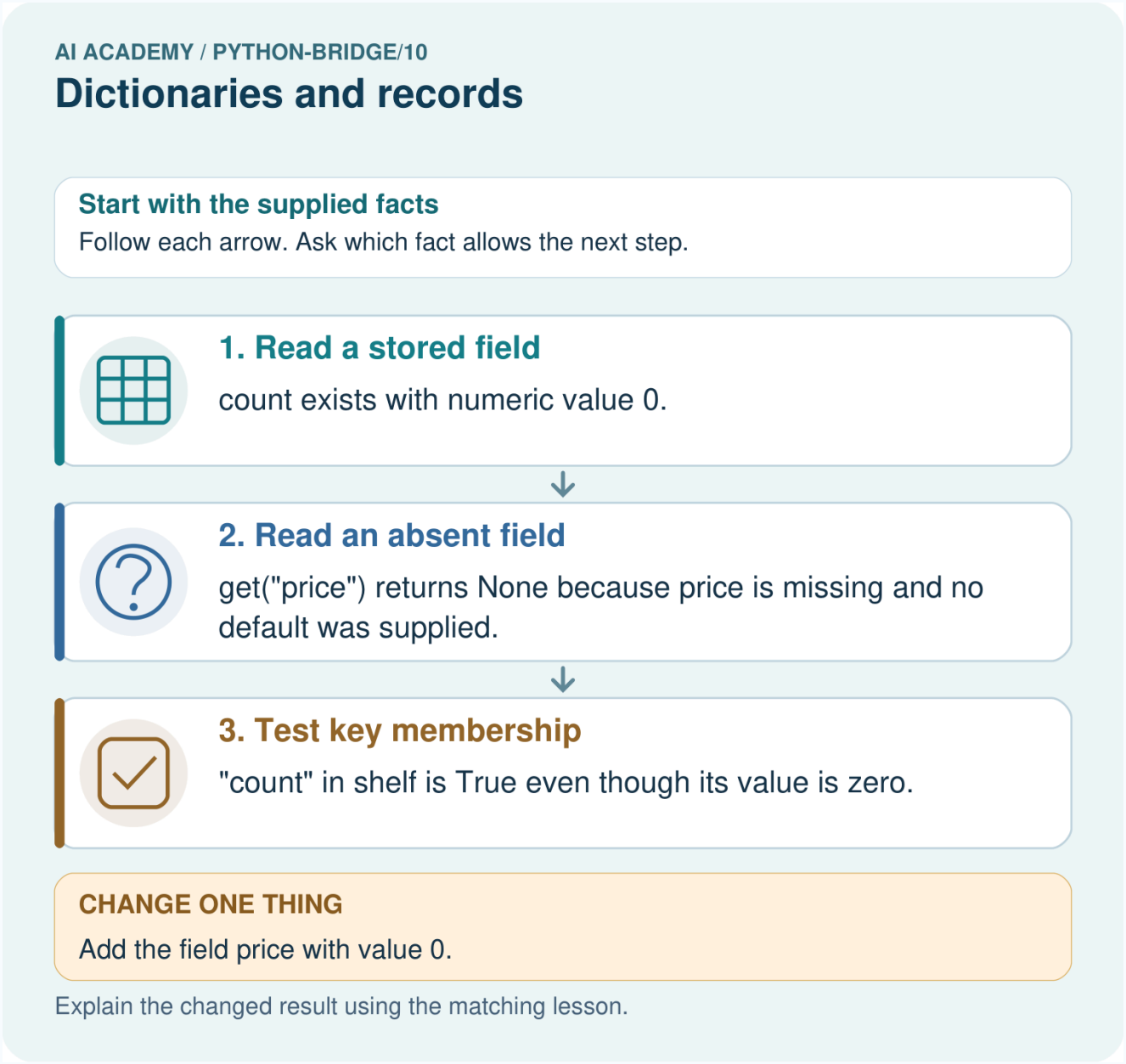
> **Set used to 12. Predict the returned spaces. Does a used value of 0 mean the field is missing?**

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Dictionaries and records

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

MAIN LESSON · SECTION 3

Guide the trace, then fade the help

> Collect predictions before execution or feedback.

Main completion task and key

Completion task: Change the Red **record** used to 9. State remaining spaces and whether capacity changes.

Teacher check: Spaces 0; capacity remains 9.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Dictionaries and **record**s. Proposed explanation: "A successful run proves the conclusion". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Named fields reduce position mistakes but do not prove that a **record** is accurate. A capacity of 2 with used of 5 is representable in a **dictionary** and violates this schema.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial question and changed-case key

Predict all three original **output**s. Explain the difference between count=0 and missing price, then predict the lookup after price=0 is added.

Outputs are 0, None, True. count is present with zero; price is absent. Adding price=0 makes get("price") return 0 without changing count membership.

Changed-case reasoning: The price lookup now returns 0 instead of None. A **record**ed zero is a present value, whereas the earlier lookup had no field.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

▼ Support and alternative plan

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and explain the purpose in the opening section.
7 to 20 minutes	Read the worked example and diagram; ask learners to trace each step aloud.
20 to 33 minutes	Complete W1 and W2, then compare predictions with observed results.
33 to 45 minutes	Collect the guided prediction before running; discuss a mismatch and its cause.
Session 2 start to 8 minutes	Retrieve the main rule without showing the previous answer.
8 to 20 minutes	Use W3 to diagnose and repair the stated failure; rerun an earlier case.
20 to 35 minutes	Collect W4 independently, including code or plan, predictions and explanations, before hints.

Time or stage	Teacher action
35 to 45 minutes	Discuss W5 and the exit; use the fresh retry when support was needed.

Use the worked example in Sections 3 and 6. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Use key cards and value cards. Extension: add a location text field and show why it does not change the remaining calculation.

▼ **Earlier wording, alternatives and picture transcripts**

Compact alternative: this two-session sequence covers selected core tasks. Use the three-session weekly plan for the full lesson, independent transfer, feedback and project work; allow extra time for extensions.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Preserve expected, observed and paper-trace records.

Supplied case materials and checked guidance

Red W2 spaces 0 and capacity 9 unchanged. Wrong-case lookup KeyError, repair key rather than add new Used field. Zero remains a measurement; absent field is not default 0. Pictured outputs 0,None,True; adding price 0 makes lookup 0. Explicit None and absence may both yield None from get but differ in membership. Selected outputs 0,3,missing; if not count misclassifies initial 0 as missing.

No arbitrary-input validation or physical capacity verification is claimed.

Diagnose misconceptions

A successful run proves the conclusion: Named fields reduce position mistakes but do not prove that a **record** is accurate. A capacity of 2 with used of 5 is representable in a **dictionary** and violates this schema.

Matching the example is enough: Require the new W4 case, explicit **prediction**s and an explanation. Fresh retry: create a Gold **record** with capacity 11 and used 3; return 8, update used to 11 and return 0.

Require field lookups rather than copied literal arguments.

A convenient shortcut preserves meaning: cupboard["Used"] raises **KeyError** here because keys are case sensitive. Correct it to "used".

Adding a new field called "Used" would create a second key and leave the original unchanged; it would not repair the intended lookup.

Use Section 9 for the **evidence** ledger. Add a boundary or missing-**input** case and **record** the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows	Next teaching action
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected bridge transfer task

> If a learner has seen this exact case, use a fresh parallel input and record exposure/support.

Selected independent task and checked key

Selected case: Predict the three **output**s and trace the named-field update. Explain why zero and a missing count produce different results.

If the condition becomes if not count, which supplied **record** is first handled incorrectly?

```
def describe(record):
    count = record.get("count")
    if count is None:
        return "missing"
    return count
record = {"count": 0}
print(describe(record))
record["count"] = 3
print(describe(record))
print(describe({}))
```

Outputs are0,3,missing. A stored0 is a supplied value; an absent field returns None under get here. The update replaces count with3 before the second call.

Using if not count would incorrectly return missing for the original count0 because zero is false in a Boolean test; testing None keeps absence separate.

Judge trace and support

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly **record**ed model failure if the learner correctly explains its **evidence** and limits.

Give one repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Identify fields

For {"name": "Red", "capacity": 9, "used": 2}, state the keys, value of used and remaining spaces.

Keys name, capacity, used; used is 2; remaining spaces 7.

W2 Trace an update

Change the Red **record** used to 9. State remaining spaces and whether capacity changes.

Spaces 0; capacity remains 9.

W3 Repair a lookup

The code reads **record**["Used"] from the Red **record**. Explain the error and repair it without adding a field.

KeyError because Used is absent. Use **record**["used"].

W5 Check the contract

A **record** has capacity 3 and used 4. It gives -1 spaces without an error. Should it be accepted? Explain.

No. The used count exceeds capacity. Successful arithmetic does not validate the **record**.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Spaces are 0. A present used field with value 0 is different from an absent field. Exit: No. The used count exceeds capacity. Successful arithmetic does not validate the **record**.

W1 Identify fields

W2 Trace an update

W3 Repair a lookup

W4 Build a record independently

Create a Green **record** with capacity 15 and used 6. Pass its fields to remaining and store the result. Update used to 10 and repeat. Predict both results.

Use a **dictionary** with name Green, capacity 15, used 6; results are 9 and then 5. The second call must read the updated field.

W5 Check the contract

Independent assessment and fresh retry

Independent W4 **evidence**: Use a **dictionary** with name Green, capacity 15, used 6; results are 9 and then 5. The second call must read the updated field. **Record** whether the learner needed no help, a hint or explanation.

Fresh retry: create a Gold **record** with capacity 11 and used 3; return 8, update used to 11 and return 0. Require field lookups rather than copied literal arguments.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Keep the portfolio, support and delayed check

> Track code execution separately from a correct paper trace.

Project evidence

Add a labelled dictionaries and **record**s **evidence** sheet to Python readiness portfolio. Show how you can read and update named **dictionary** fields.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this **record** to last week's **record**; a matching answer without a trace is insufficient.

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Support and selected assessment guidance

Use key cards and value cards. Extension: add a location text field and show why it does not change the remaining calculation.

Selected independent answer: Outputs are0,3,missing. A stored0 is a supplied value; an absent field returns None under get here. The update replaces count with3 before the second call. Using if not count would incorrectly return missing for the original count0 because zero is false in a Boolean test; testing None keeps absence separate.

Keep earlier W4 transfer as additional practice; do not substitute its numbers or key for this selected task. Record prior exposure and use a fresh parallel case after support.

Delayed return

Suggested return points in this level: python-bridge/11. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and **record** support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	

Record	Teacher evidence
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to the next topic

Next week: Reading structured data. Reuse this week's explicit contracts and **evidence**.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route without repeating the teaching cycle.

Planning and assessment reference

Teaching target: Read and update named **dictionary** fields

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Resources:** Sections 1–6; record the brief recall in the workbook.

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> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: Change the Red **record** used to 9. State remaining spaces and whether capacity changes.

Reasoned practice answer: Spaces 0; capacity remains 9.

Check every learner before discussion: Use the worked case for Dictionaries and **record**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Named fields reduce position mistakes but do not prove that a **record** is accurate. A capacity of 2 with used of 5 is representable in a **dictionary** and violates this schema.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Add the field price with value 0. Require a **prediction** and a reason before comparing. Expected result: The price lookup now returns 0 instead of None.

Reason: A **record**ed zero is a present value, whereas the earlier lookup had no field.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. **Record** support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Python readiness portfolio. Keep runnable code, **prediction**s, actual **output**s, one repaired error and an explanation of a function on changed **input**s.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity **record**; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback and evidence records**

REFERENCE · SECTION 8

Reference · Feedback and evidence records

> Keep individual first attempts separate from supported rehearsal.

Feedback and repair

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. **Record** support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

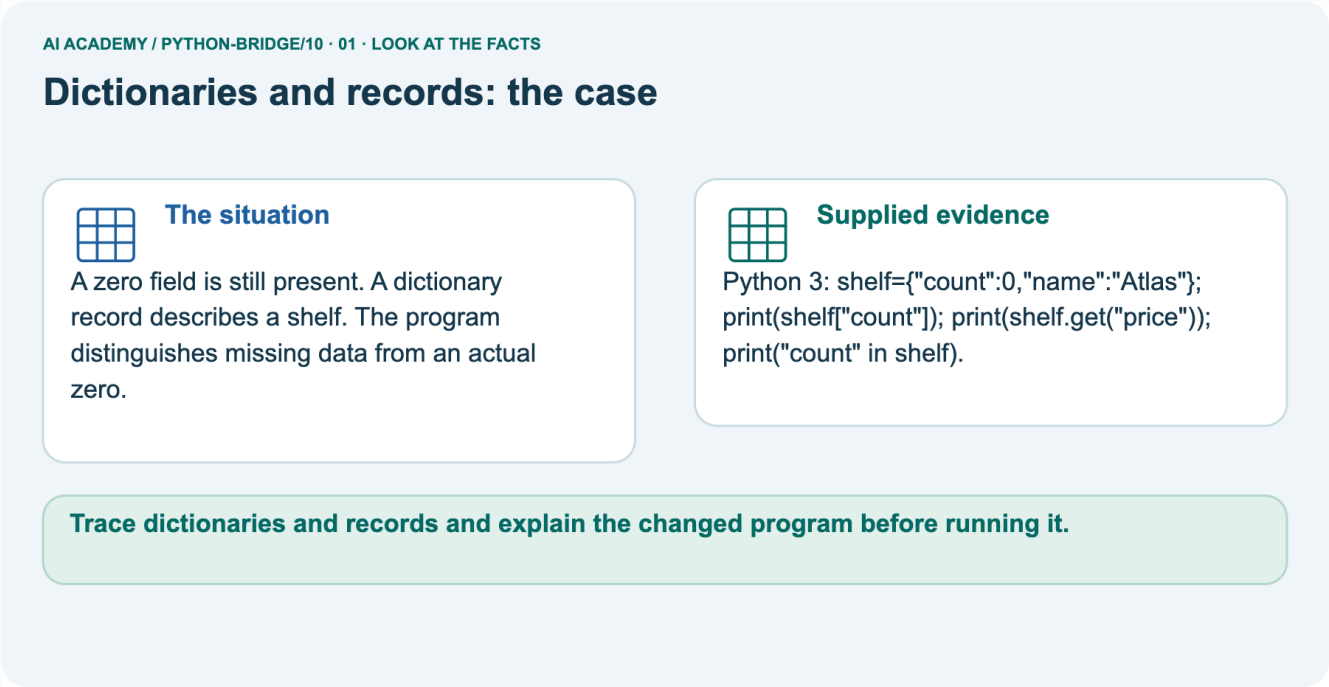
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Use the relevant picture for a specific reasoning move.

01. Dictionaries and records: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

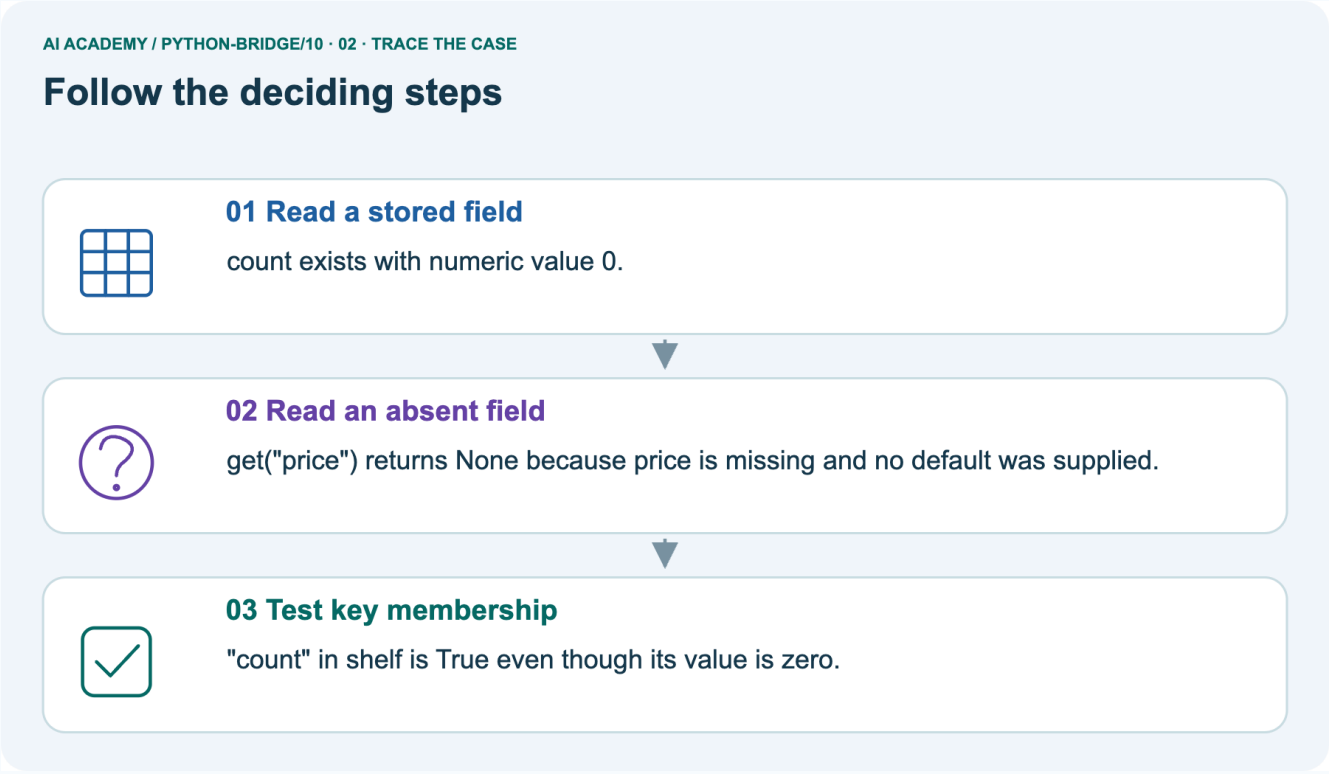
A zero field is still present. A **dictionary record** describes a shelf. The program distinguishes missing data from an actual zero.

Supplied facts: Python 3: shelf={"count":0,"name":"Atlas"}; print(shelf["count"]); print(shelf.get("price")); print("count" in shelf).

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



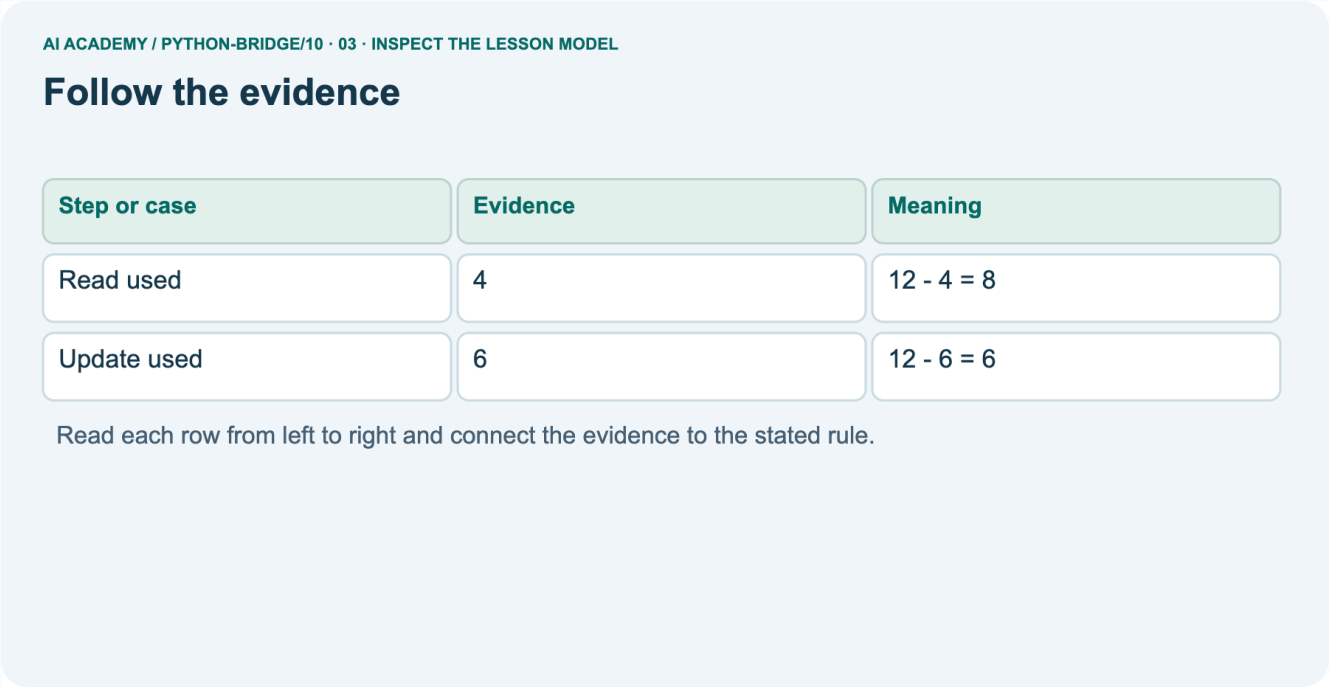
Follow the labelled picture with the matching explanation. Open the vector diagram

Read a stored field: count exists with numeric value 0. → Read an absent field: get("price") returns None because price is missing and no default was supplied. → Test key membership: "count" in shelf is True even though its value is zero.

Point to each numbered stage and explain which **record**ed **input** it uses.

Teacher prompt: Point to each numbered stage and explain which **record**ed **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Follow the evidence



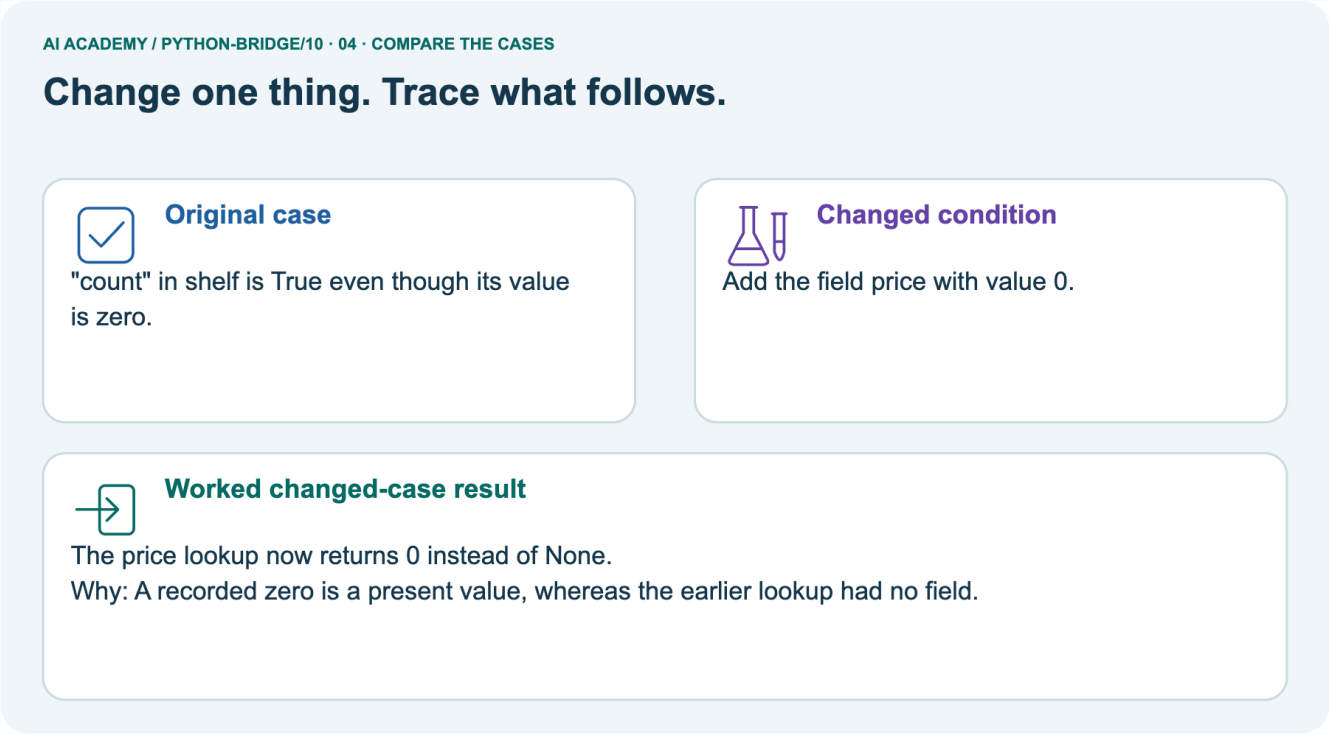
Follow the labelled picture with the matching explanation. Open the vector diagram

Follow the **evidence**. Read used | 4 | 12 - 4 = 8. Update used | 6 | 12 - 6 = 6

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: "count" in shelf is True even though its value is zero. Changed condition: Add the field price with value 0. Expected result: The price lookup now returns 0 instead of None.

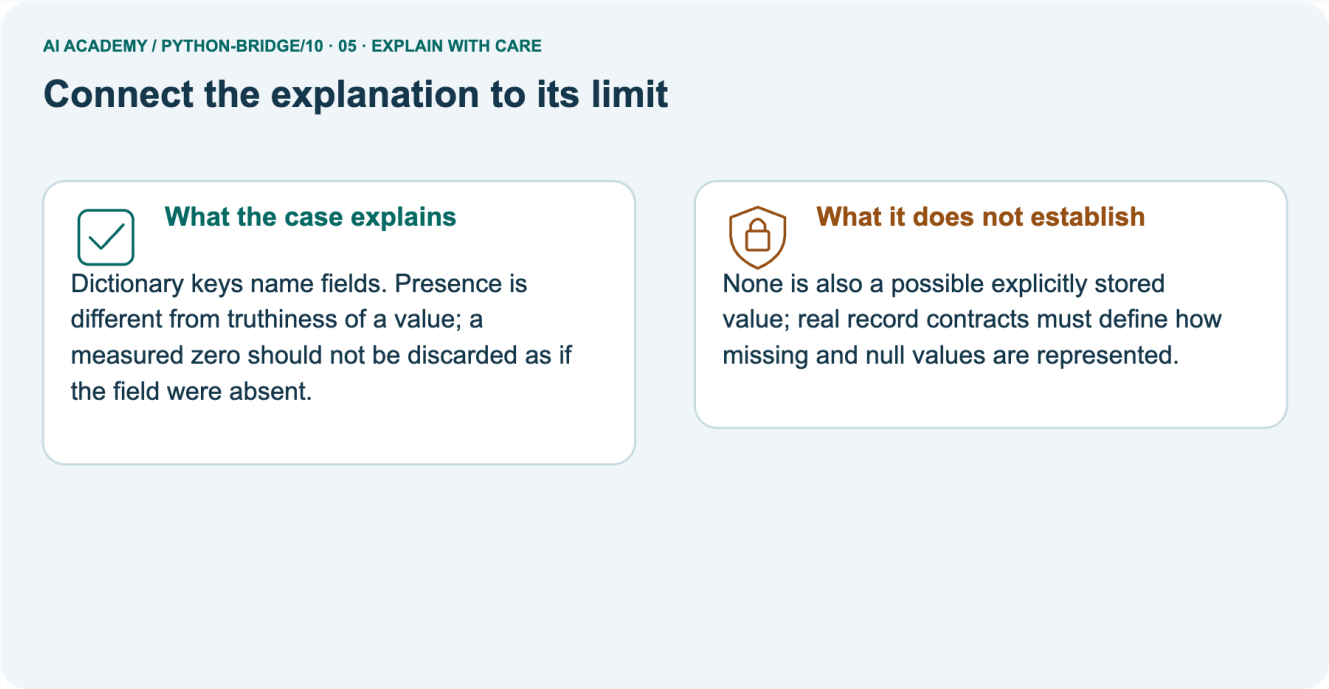
Reason: A **record**ed zero is a present value, whereas the earlier lookup had no field.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: A **record**ed zero is a present value, whereas the earlier lookup had no field.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Dictionary keys name fields. Presence is different from truthiness of a value; a measured zero should not be discarded as if the field were absent.

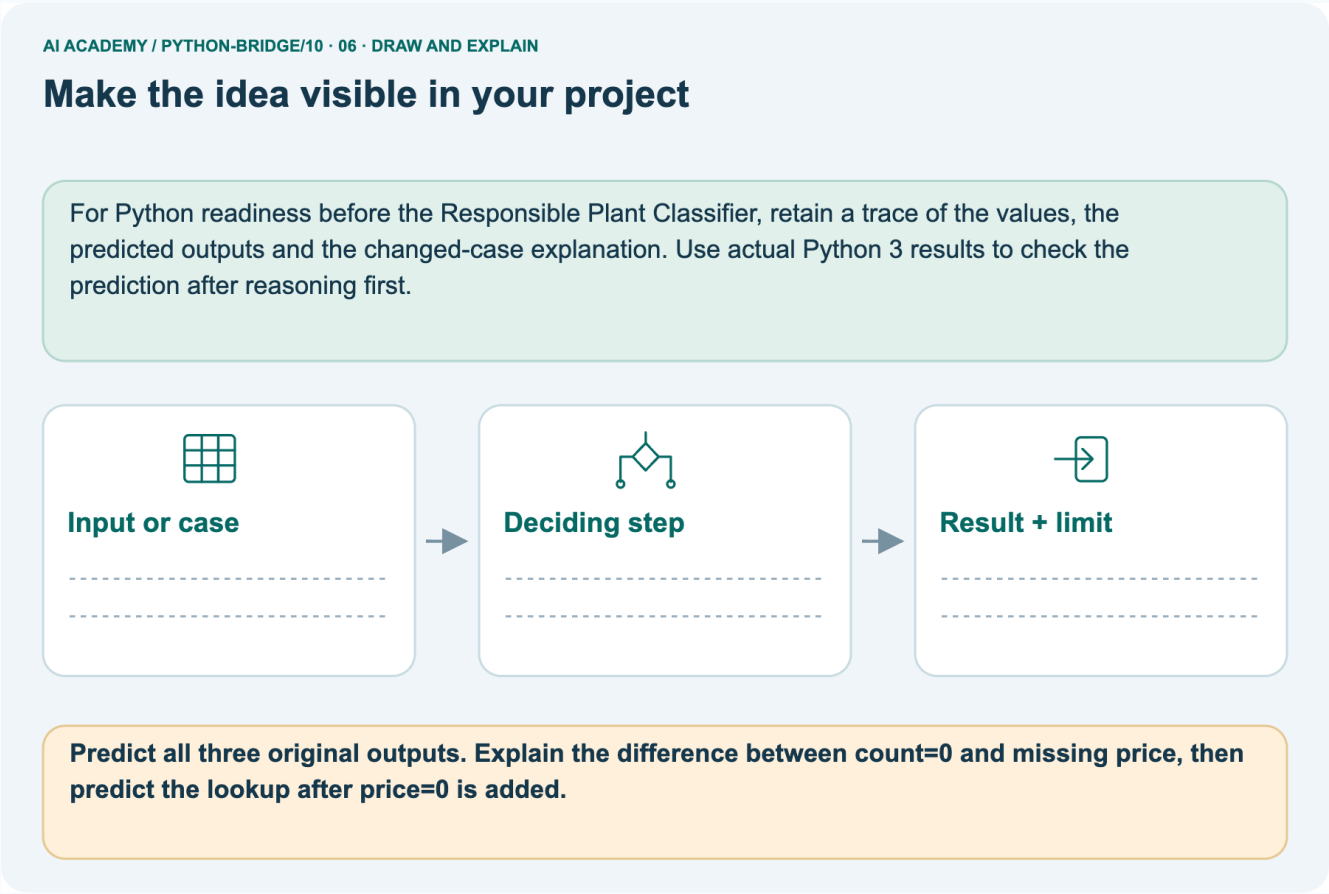
Limit: None is also a possible explicitly stored value; real **record** contracts must define how missing and null values are represented.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: None is also a possible explicitly stored value; real **record** contracts must define how missing and null values are represented.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Draw your **input**, deciding step, result and limit. Supported question: Predict all three original **output**s. Explain the difference between count=0 and missing price, then predict the lookup after price=0 is added.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: **Output**s are 0, None, True. count is present with zero; price is absent. Adding price=0 makes get("price") return 0 without changing count membership.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Dictionaries and records: the case

02. Follow the deciding steps

03. Follow the evidence

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.